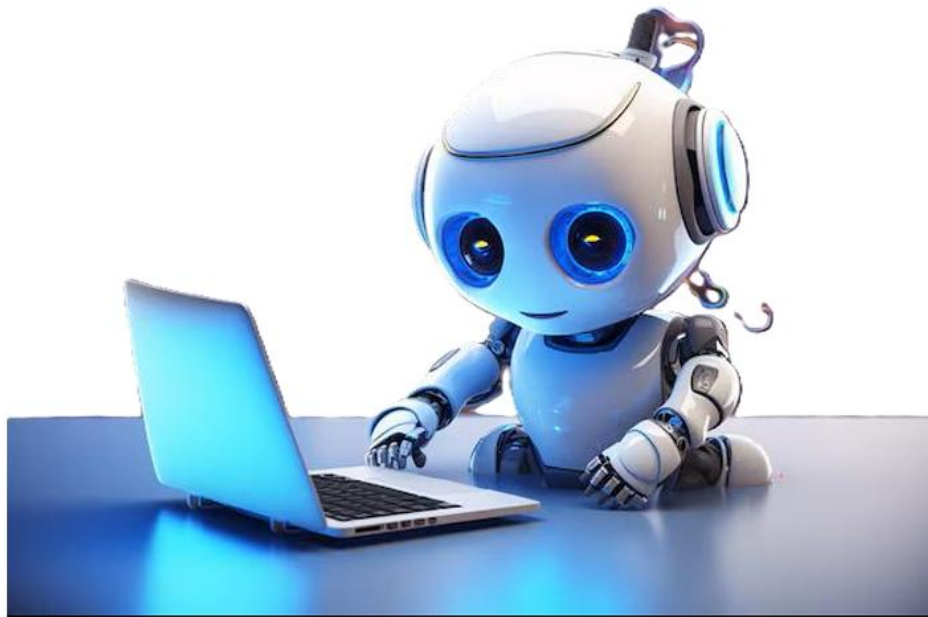


Implementation Plan



AI Proposal Generator

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1. Project Overview

The **AI Proposal Generator** is a web-based system designed to automatically generate structured project proposals based on client requirements entered by internal staff.

The system aims to:

- Reduce manual effort in proposal creation
- Improve response time to client inquiries
- Ensure consistency and professionalism in proposals

2. Requirements Overview

2.1 Functional Requirements

Input Module

Users can enter:

- Client Name
- Company Name
- Project Title
- Project Description
- Required Features
- Budget Range (optional)
- Timeline (optional)
- Platform Type

AI Proposal Generation

System should:

- Analyze user input
- Generate structured proposal including:
 - Introduction
 - Requirement Understanding
 - Proposed Solution
 - Features List
 - Timeline
 - Budget
 - Conclusion

Output Module

- Display proposal in structured format
- Allow:
 - Editing
 - Copying
 - PDF download (optional)

Future Scope (Optional)

- Save proposals
- Search proposals
- Proposal history

2.2 Non-Functional Requirements

- Simple and user-friendly UI
- Fast response time
- Basic validation
- High readability of output

2.3 Out of Scope

- Authentication system
- Advanced AI training
- Complex pricing algorithms
- Full CRM integration

3. System Architecture

The system follows a three-tier architecture: Frontend (UI), Backend (processing & API), and AI Service Layer.

1. Frontend (Client Layer)

- User interface for input and output
- Handles form validation and display

2. Backend (Application Layer)

- Processes input data
- Communicates with AI API
- Formats response

3. AI Service Layer

- External AI API
- Generates proposal content

4. Architecture Flow

- User enters client details in the frontend
- Frontend validates inputs
- Request sent to backend API
- Backend processes and sends prompt to AI service
- AI generates proposal content
- Backend formats response
- Response sent to frontend
- Proposal displayed to user

5. Technology Stack

Frontend

- React.js
- Tailwind CSS
- Axios (API calls)

Backend

- Node.js (Express.js)

AI Integration

- OpenAI API (or equivalent)

Optional Tools

- PDF Generation: jsPDF
- Version Control: Git
- Testing Tools: Postman

6. Development Approach & Methodology

Phase 1: Requirement Analysis & Planning

- Review requirements
- Define workflow
- Identify test scenarios
- Finalize tech stack

Phase 2: UI Development

- Build input form
- Design output layout
- Add validations

Phase 3: Backend Development

- Setup API
- Integrate AI service
- Handle responses

Phase 4: Integration

- Connect frontend and backend
- Validate full workflow

Phase 5: Feature Enhancements

- Edit functionality
- Copy feature
- PDF export

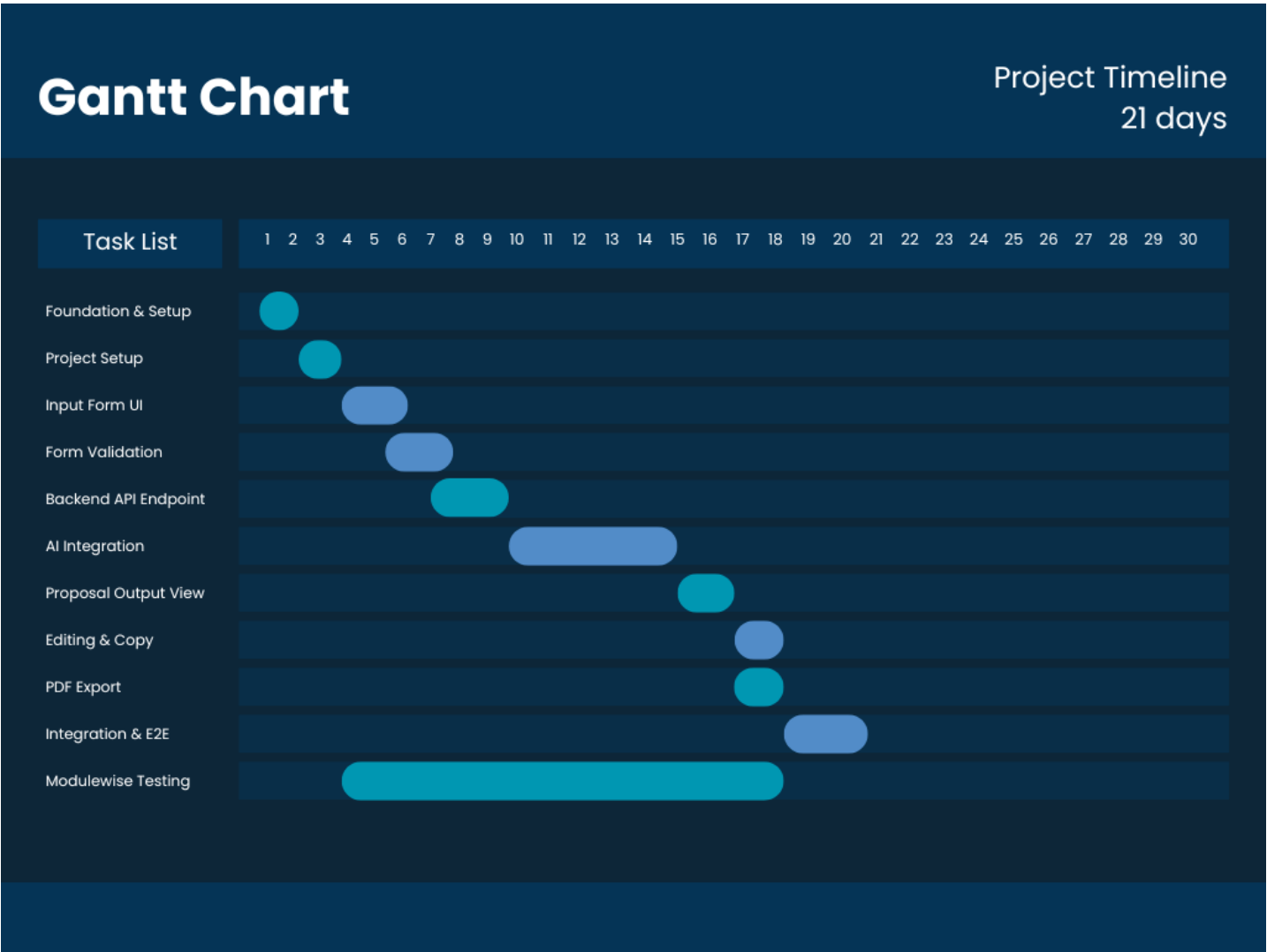
Phase 6: Testing & QA

- Functional testing
- UI testing
- API testing
- Bug fixing

Phase 7: Deployment & Documentation

- Prepare final build
- Create documentation
- Provide sample outputs

7. Project Timeline



#	Module	What it covers	Dependency
M1	Project Setup	Repo, env config, folder structure	None
M2	Input Form UI	All 8 fields, layout, responsive design	M1
M3	Form Validation	Required field checks, input sanitization, error messages	M2
M4	Backend API Endpoint	Receives form payload, basic routing, error handling, API key test	M1
M5	AI Integration	Prompt engineering, API call, response parsing	M4
M6	Proposal Output View	Section-by-section rendering of AI response	M5
M7	Editing & Copy	Inline editing of sections	M6
M8	PDF Export	Generate downloadable PDF from proposal	M7
M9	Integration & E2E	Full flow wiring, edge cases, performance check	All

Parallel Dev + QA tracks

Week	Dev track	QA track
1	M1, M2	Test plan, M2 test cases (form fields, layout)
2	M3, M4	Execute M2 tests, M3+M4 test cases (validation rules, API responses)
3	M5, M6	Execute M3+M4 tests, M5+M6 test cases (AI output quality, rendering)
4	M7, M8	Execute M5+M6 tests, M7+M8 test cases (edit, copy, PDF accuracy)
5	M9	Execute M7+M8 tests, full integration + UAT + regression

8. Task Breakdown (Modules)

Module 1: Input Module

- Form UI
- Validation logic

Module 2: AI Processing Module

- API endpoint
- Prompt engineering
- AI response handling

Module 3: Output Module

- Structured display
- Editable fields

Module 4: Utility Features

- Copy to clipboard
- PDF export

Module 5: Future Module

- Save and manage proposals

9. Testing Strategy & QA Plan

9.1 Testing Types

Functional Testing

- Verify all features work correctly

UI Testing

- Layout, responsiveness, readability

API Testing

- Request/response validation

Integration Testing

- End-to-end workflow

Negative Testing

- Invalid inputs
- Empty fields

Performance Testing

- AI response time

9.2 QA Deliverables

- Test Plan
- Test Scenarios
- Test Cases
- Bug Reports
- Test Execution Report
- Test Summary Report

10. Risk Identification & Mitigation

Risk	Impact	Mitigation
Prompt quality takes iteration	Proposals may be generic, inaccurate, or poorly structured which directly undermining the product's core value.	Reserve 2 full days in Week 3 specifically for prompt tuning. Set acceptance criteria for output quality before moving on.
Scope creep from stakeholders	New feature requests (email sending, CRM integration, templates) mid-development delay delivery	Strictly follow the "Out of Scope" section in the requirements document. Log new requests in a backlog for future sprints.

Non-technical user adoption	If UI is not intuitive, business/sales staff may avoid using the tool — defeating the time-saving goal.	Run usability review with 1–2 non-technical team members during Phase 3. Prioritize simple language, tooltips, and clear CTAs.
UAT feedback arrives late	Last-minute fixes rushed without proper retesting can break things that were already working. You end up introducing new bugs while closing UAT ones.	Get stakeholder commitment to respond within 2 business days. Send UAT scenarios 2 days before Week 5 starts so they can prepare.

11. Assumptions

- Basic proposal format is acceptable.
- AI API is available and reliable.
- UI wireframes are approved.
- Users are internal staff.
- Hosting infrastructure is arranged in parallel.

12. Dependencies

- AI API service
- Internet connection
- Development tools
- PDF library (optional feature)

13. Deliverables

- Working web application
- Source code
- Documentation
- Sample proposals.

14. Success Criteria

- Proposal generated based on input
- Output is structured and readable
- Reduces manual effort significantly
- Minimal critical defects